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Mechanisms for Prediction Markets

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Crypto's speculative nature is its greatest strength and most glaring weakness. On the one hand, it attracts fresh waves of capital to fuel the technology's development and secure the underlying economic protocols. On the other hand, the volatile booms and busts distract from the technology's underlying utility and lead many to conclude that permissionless blockchains are nothing more than unregulated casinos. The recent surge in onchain prediction markets exemplifies this duality perfectly. These platforms harness crypto's speculative nature to power

an engine to predict the future with unprecedented accuracy. In doing so, prediction markets have emerged as one of crypto's most compelling and concrete use cases to date.

Prediction markets are nothing new. The earliest known forms around 500 years ago involved political betting on papal successors, while the first instance on Wall Street occurred around 1884. In the digital age, the Iowa Electronic Market (IEM), Predictit, and Manifold have had some traction as niche online platforms. Since the advent of smart contracts, onchain prediction markets have been pioneered by projects like Augur, Gnosis, and Omen, and recently started to have breakout success with Polymarket. Onchain prediction markets have grown more than 5x in volume since August 2023, with a current TVL of ~$130 million and $1 billion in cumulative volume since the start of 2024.

In this article, we will dive into the mechanisms behind onchain prediction markets to better understand why they exist, how they work (specifically Polymarket), and how they might be improved.

Understanding Prediction Markets

Prediction markets allow people to directly bet on the outcomes of future events. By betting on the probability of some outcome, the private beliefs of participants are aggregated to indicate public sentiment regarding the likelihood of the event.

A prediction market can take one of three forms: binary, categorical, or continuous:

Binary markets allow participants to bet YES/NO on an outcome. In a binary market, the perception of the likelihood of an event happening is expressed as a probability where the likelihoods of the positive and negative options are positive values that sum to 1.

- Positive Outcome (Yes): A where A < 1
- Negative Outcome (No): B where B = 1-A

Categorical markets operate similarly, but instead of just YES/NO, participants can bet on a set of multiple predetermined outcomes. Again, each option is expressed as its probability where each value is positive and all options sum to 1.

- Outcome A: A where A = 1 - B - C
- Outcome B: B where B = 1- A - C
- Outcome C: C where C = 1 - A - B

Continuous markets account for many values across a range indicated by the participants' predetermined constraints. These markets typically operate until a set expiration date, determining winners based on whether the outcome falls within the specified range.

- $Asset >= X on a set date
- Y < $Asset < X on a set date.

A common misconception is that prediction markets are more vulnerable to manipulation by wealthy participants than traditional forecasting institutions like weather services, government agencies, or public opinion polling. However, this isn't typically the case. Manipulators generally do not significantly impact information aggregation or prices due to the presence of noise traders—participants who trade for reasons unrelated to market information, such as errors or insurance needs. This random trading activity dilutes the impact of manipulative trades, preventing them from notably altering market trends. Additionally, rational traders often increase their trading activity in response to this noise, which can paradoxically enhance market accuracy and efficiency (see Robin Hanson 2004 and 2005).

From Traditional to Onchain Prediction Markets

Before prediction markets, traders would bet on future outcomes indirectly through derivatives markets. Derivatives markets use derivatives contracts such as futures, options, or swaps to bet on the future price of an asset indirectly related to that outcome. Let's say a trader believes there will be a drought shortly. To profit off this prediction, the trader purchases a futures contract for wheat, locking in the lower price now and anticipating higher prices in the future. If the drought occurs and the price of wheat increases, the trader can activate their contract and arbitrage the commodity at the higher spot price for profit—a much more complex process than simply betting on the drought.

While derivatives markets have historically facilitated indirect bets on outcomes through complex financial instruments, prediction markets offer a more direct and streamlined approach. Typically, traditional ones use a **Continuous Double Auction (CDA)** design. CDAs operate like an order book, where a centralized intermediary known as an operator matches buy orders and sell orders. This makes traders counterparties to each other and relies on market forces to incentivize traders to converge on accurate prices. CDAs are implemented in popular prediction markets such as the IEM and Predictit.

Traditional prediction markets face several limitations (see Vitalik 2021):

- **Single Point of Failure**: Centralized systems are vulnerable to technical failures, cyberattacks, or operational issues that can disrupt market activity. If the central entity goes offline, the entire market is affected.
- **Bias & Manipulation**: Users must trust a single entity to maintain fairness and transparency. There is a risk of manipulation or bias by

the central authority.

- **Lack of Privacy**: Participants in centralized markets often need to provide personal information, which can deter those who value privacy or are concerned about data breaches.
- **High Costs**: Centralized platforms often charge significant fees for participating in the market. These fees can include transaction costs, withdrawal fees, and other charges that eat into potential profits.
- **Limited Access**: Many centralized prediction markets restrict participation based on geographic location, limiting their accessibility. This can prevent a truly global and diverse set of opinions from being reflected in the market prices.
- **Limited Range of Events**: Centralized markets may offer a limited range of events for betting, often focusing on popular topics. This restricts the ability of participants to bet on niche or emerging events.
- **Low Limits**: Many centralized prediction markets impose low betting limits on participants, which can stifle serious trading and reduce users' ability to express strong confidence in a particular outcome.

These limitations reduce liquidity and participation, ultimately restricting their reach and accuracy. In contrast, onchain prediction markets utilize smart contracts, eliminating the need for trusted infrastructure and intermediaries, all carrying costs typically passed on to users. These protocols can operate autonomously, providing deterministic, transparent results and a highly accessible trading environment. Users can place bets of any amount at any time with minimal effort and reduced costs.

Onchain prediction markets usually rely on Automated Market Makers (AMM), a blockchain-based alternative to CDA that allows liquidity pools to act as counterparties to all trades. In this design, pools managed by

smart contracts automatically hold and distribute liquidity to traders. These pools maintain accurate prices by relying on an algorithm continuously adjusting pool quantities based on supply and demand.

However, AMMs come with their own set of challenges, including:

- **Slippage**: Slippage occurs when there is a difference between the expected price of a trade and the price at which the trade is executed. In AMMs, slippage tends to increase with the size of the trade, particularly in pools with lower liquidity. This means larger orders can significantly move the price, potentially leading to less favorable trade execution.
- **Speed**: Transaction speeds in AMMs can be affected by the underlying blockchain's performance. During times of high demand, the network may become congested, leading to slower transaction confirmations. This delay can cause the prices on the AMM to become outdated compared to fast-moving markets, impacting trading strategies.
- **Impermanent Loss**: Liquidity providers in an AMM face this risk due to volatility in the price of assets within the pool. When the price of pooled assets changes after they are deposited, the proportional value of their stake in the pool changes compared to holding the assets outside the pool. This can result in a financial loss if the liquidity provider decides to withdraw their assets during a period of unfavorable price movement.
- **MEV**: AMMs can be vulnerable to Maximal Extractable Value (MEV), where validators prioritize, include, or exclude transactions to maximize profits. This can lead to front-running, where miners execute their trades before processing others already in the queue, taking advantage of price movements that are about to occur due to pending transactions. This undermines the fairness and efficiency of

the market, potentially leading to losses for regular traders.

Despite these challenges, onchain prediction markets have seen a significant rise in recent months, with Polymarket becoming a leading indicator of sentiment surrounding the outcome of the US presidential election, according to the Wall Street Journal. Currently, Polymarket makes up about 75% of the total value currently locked in on-chain prediction markets. Below, we'll be able to explore its workings further.

What is Polymarket and How Does it Work?

Polymarket, launched in 2020, is a non-custodial prediction market that

runs on the Polygon blockchain. The platform provides users with diverse binary outcomes to bet on, known as 'information markets.' In these markets, users wager with the stablecoin USDC. By supplying USDC as collateral, users obtain outcome tokens representing either a market's positive or negative outcome. At the market's conclusion, each outcome token on the winning side becomes redeemable for 1 USDC, while the outcome tokens of the losing side become worthless. In essence, the winners receive the collateral from the losers.

See the example below:

- Binary market
    - Token Y (Positive- Yes): $0.65 per share
    - Token N (Negative- No): $0.35 per share
- Outcome does occur, Market concludes
    - Positive bettors win
- Payouts
    - All Y tokens increase to $1
        - Winners earn $0.35 on each share
    - All N tokens fall to $0
        - Losers lose $0.35 on every share

The more likely an outcome is, the lower the expected payout, and vice versa. This dynamic creates a unique environment where the invisible hand of economics drives markets to converge on accurate prices. If an outcome is likely grossly overpriced or underpriced, a trader has a significant financial incentive to bet against or for it and turn a profit. This

accuracy increases with more users and liquidity.

In its current design, the Polymarket team creates and deploys all markets on the platform. The Polymarket community suggests, discusses, and reviews new markets through the Polymarket Discord channel. This allows the Polymarket community's preferences to be reflected in the markets available on the platform. Below, we will explore some of its core mechanism implementations.

Conditional Tokens

Developed by Gnosis, the Conditional Token Framework (CTF) is a sophisticated way to create, manage, and settle bets on future outcomes.

- **Token Standard**: It uses the ERC1155 standard, which allows for more efficient and flexible handling of multiple types of outcome tokens within a single contract and reduces gas costs compared to individual ERC20 tokens.
- **Customization**: The framework supports creating markets with various conditions and outcomes, making it adaptable to different prediction markets. It also allows for integration with various Oracle solutions.

Polymarket's CTF implementation allows for the following operations:

- **Splitting**: Users can split one unit of collateral (USDC) into one unit of each outcome token, creating liquidity in the market.
- **Merging**: Users can merge one unit of each outcome token back into one unit of collateral. This allows for arbitrage opportunities and helps maintain price equilibrium.

- **Redeeming**: Users can redeem their winning outcome tokens for the underlying collateral after a market resolves.

Central Limit Order Book (CLOB)

The CLOB is Polymarket's default system, which takes inspiration from the traditional Continuous Double Auction (CDA) but adapts it to work onchain. This mechanism operates with these key features:

- **Order Matching**: Orders to buy or sell outcome tokens are organized in the CLOB, where buy orders are matched with the lowest available sell orders and sell orders with the highest available buy orders, based on price priority. Trades are executed at the best available price, with earlier orders receiving priority when prices are equal.
- **Hybrid On-chain/Off-chain System**: Although order matching is done off-chain by Polymarket, the settlement (i.e., the actual exchange of tokens) is executed onchain. The exchange contract autonomously settles trades, tracks ownership, handles payouts, and issues/burns outcome tokens in exchange for USDC collateral.
- **Order Types and Participant Roles**:
    - A 'limit order' is an instruction to buy or sell an outcome token at a specified price or better, which will only execute when the market price meets the price set by the trader. These orders enter the order book and wait for the market to reach their specified price. Traders who place limit orders are called 'makers.' Makers provide liquidity to the market, essentially 'making' the market for specific prices.
    - A 'market order' is an instruction to buy or sell an outcome token at the best available current price. Traders placing market orders are called 'takers.' Takers provide the necessary market

movement by executing trades, which helps discover the price and maintain market fluidity.

Fixed Product Market Maker (FPMM)

Also developed by Gnosis, the FPMM serves as an alternative to the CLOB, acting as a fallback mechanism in times of insufficient liquidity or order volume in the CLOB. It is an automated market maker (AMM) specific to prediction markets, which operates in the following way:

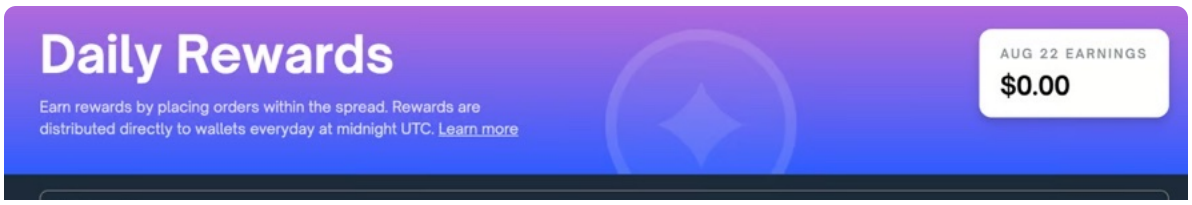
- **Liquidity Pools**: Each market has a liquidity pool comprising tokens representing possible outcomes. These pools facilitate trading by ensuring there is always a counterparty to take the opposite side of a bet.
- **Price Adjustment**: Prices within the liquidity pool automatically adjust based on trades, maintaining a constant product of the quantities of the two different tokens involved. This ensures that prices react dynamically to changes in supply and demand, stabilizing the market and encouraging balanced trading activity.
- **No Operator**: This mechanism does not require an active intermediary to match trades. The liquidity pool itself acts as the counterparty to all trades, adjusting prices algorithmically based on supply and demand.

Liquidity Rewards Program

Polymarket continuously runs a liquidity rewards program to reduce trading slippage within markets. This enables more participation, ultimately increasing the accuracy of predictions. Unlike typical liquidity mining programs incentivizing AMM liquidity, Polymarket adapts a design

by dYdX and Blur to incentivize order book liquidity on its CLOB.

- **Daily Rewards**: Anyone can earn money by placing limit orders on particular markets within a predefined spread that the Polymarket team sets to keep specific markets active and balanced. Rewards per market per day are fixed in USDC and are paid out to eligible Makers daily at midnight UTC based on their score relative to the other Makers in that market.
- **Scoring System**: The program assesses the quality of liquidity provided by market makers based on factors like order size, duration, and proximity to the market's current price, rewarding the best contributions to market health. The methodology adapts dYdX's scoring mechanism, which is adjusted for binary contract markets.
- **Leaderboard Competitions:** Polymarket plans to further incentivize liquidity through leaderboard competitions, which will likely provide an extra reward to the top Makers across all markets daily.
- **Rewards Funding**: The funding for the rewards in Polymarket's Liquidity Rewards Program primarily comes from UMA tokens provided by the UMA DAO from the UMA treasury. These rewards are intended to stimulate trading activity and liquidity provision in markets that use UMA's Oracle services. In March 2023, Polymarket requested 1,250,000 $UMA from the UMA DAO to allocate towards growing Polymarket's new Order Book. The reliance on UMA tokens for Polymarket's liquidity rewards began in 2022. In the 2023 forum post, Polymarket team members expressed that the platform may consider incorporating trading fees in the future, from which the rewards will be funded.



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Market	Spread	Min share	Daily reward	Ends	Earnings	Percent	Price
Trump mentioned fewer than 20 times at DNC on day 3?	±3¢	20	40	∞	$0.00	-	Yes 24¢ No 76¢
Trump mentioned between 20 and 29 times at DNC on day 3?	±3¢	20	40	∞	$0.00	-	Yes 23.5¢ No 76.5¢
Trump mentioned between 30 and 39 times at DNC on day 3?	±3¢	20	40	∞	$0.00	-	Yes 50.5¢ No 49.5¢
Trump mentioned between 40 and 49 times at DNC on day 3?	±3¢	20	40	∞	$0.00	-	Yes 51¢ No 49¢
Trump mentioned between 50 and 59 times at DNC on day 3?	±3¢	20	40	∞	$0.00	-	Yes 51¢ No 49¢

Figure 1: Polymarket's Daily Rewards Page

UMA Oracle

Oracles are needed to resolve each market based on the real-world outcome of the corresponding event. Rather than relying on a single trusted party to feed this input to the smart contract, Polymarket leverages UMA's decentralized oracle.

This mechanism works in the following way:

- **Optimistic Oracle Model**: This model operates on the principle of "innocent until proven guilty." Participants proposing data must post a bond, initiating a review period. During this time, if other participants suspect the data is incorrect, they can challenge it by posting a counter bond.
- **Dispute Resolution**: Challenges are resolved by the UMA Data Verification Mechanism (DVM), where UMA token holders vote to determine the validity of the disputed data. If the data is affirmed, the bond is returned to the proposer with additional rewards; if it is overturned, the bond is forfeited.
- **Data Finalization**: The data is considered final and actionable after the liveness period passes without disputes or after the DVM resolves

any disputes.

Several mechanisms within the UMA Oracle and Polymarket process are designed to mitigate this risk and prevent incorrect data.

- **Bond Requirement**: The necessity for proposers to bond their submissions incentivizes the accuracy of the data, as incorrect data risks the forfeiture of the bond.
- **Tokenonomic Deterrents**: The structure of UMA's token economy makes it prohibitively expensive to influence outcomes through token-majority ownership, effectively deterring potential corruption by making the cost of dishonesty exceed possible gains.
- Bulletin Board**:** The Polymarket integration incorporates a Bulletin Board that allows market creators to directly question and communicate with the UMA Oracle regarding the accuracy and relevance of the data being fed into their markets. This direct communication channel helps ensure that the data used in market outcomes is reliable and transparent, allowing for corrections or clarifications when needed.

CLOB or FPMM? Tensions Between Short-Term and Long-Term Designs

Beyond the case of the Polymarket, choosing to rely on CLOB versus FPMM as the primary mechanism in an onchain prediction market may arise from different outcome preferences and timelines. While the CLOB can perform better by being capital efficient and highly liquid, the FPMM is less prone to manipulation due to being intermediary-less, censorship-resistant, and end-to-end verifiable. If the short-term goal is to onboard as many users as possible, resorting to mechanisms such as the CLOB may be the best option. However, in the long term, and as the onchain prediction market grows and becomes more liquid, switching to

manipulation-resistant mechanisms like FPMM may be a better choice.

Mechanisms for Addressing Insufficient Liquidity

As mentioned, liquidity remains one of the most pressing challenges of onchain prediction markets. Below, we outline potential mechanisms to address this obstacle.

Leveraged Outcome Tokens

Leveraged outcome tokens can help solve liquidity challenges in onchain prediction markets. This approach allows traders to purchase YES/NO tokens with leverage, such as 2x or 3x, by paying a premium over the standard price. The premium paid for these leveraged positions contributes directly to the market's liquidity pools, creating a unique mechanism that incentivizes liquidity provision while offering higher potential returns to traders.

BENEFITS	RISKS
• **Incentivizes Liquidity Provision**: The direct contribution of premiums to liquidity pools makes liquidity provision more attractive, potentially solving the "chicken-and-egg" problem often faced by new prediction markets. • **Attracts a Diverse Trader**	• **Rebalancing Issues**: Leveraged tokens may require frequent rebalancing depending on the implementation, which can lead to tracking errors or unexpected outcomes over more extended holding periods. • **Potential for Manipulation**: The

Base: The availability of leveraged options appeals to risk-averse and risk-seeking traders, potentially increasing overall market participation and volume.	leveraged nature of these tokens could make the market more susceptible to price manipulation by large traders or coordinated groups. • **Liquidity Concentration**: The system might inadvertently concentrate liquidity in the most popular or volatile markets, potentially leaving other markets underserved.

Protocol-Owned Liquidity (PoL)

PoL involves the prediction market protocol retaining a significant portion of the liquidity rather than relying solely on external liquidity providers. Using a part of its treasury or collected fees to fund and maintain liquidity pools directly, the platform can ensure consistent liquidity availability and respond swiftly to market demands.

BENEFITS	RISKS
• **Enhanced Market Stability**: PoL ensures that liquidity is available even during volatile periods,	• **Centralization Risks**: As the protocol controls a large portion of the liquidity, it may lead to

helping to stabilize prices and trading conditions. This can lead to more efficient markets and improved price discovery.

- **Operational Control**: The platform gains direct control over liquidity, enhancing its ability to manage the market environment effectively. This control allows for quick adjustments to liquidity levels based on market needs.
- **Reduced Reliance on External Providers**: By owning a significant portion of the liquidity, the protocol becomes less dependent on external liquidity providers, potentially reducing the impact of "mercenary capital" that might leave the platform during market downturns.
- **Sustainable Fee Structure**: With PoL, the protocol can potentially offer users more concerns about centralization and potential manipulation
- **Market Exposure**: The protocol directly exposes the market to risks, including potential losses, if it moves against its position.
- **Capital Intensity**: Implementing PoL requires significant upfront capital from the protocol, which may be challenging for newer or smaller platforms.
- **Opportunity Cost**: Capital allocated to liquidity provision cannot be used for other purposes, potentially limiting the protocol's ability to invest in development or marketing.

competitive fees, as it does not rely solely on incentivizing external liquidity providers.	

Global Liquidity Pool

A global liquidity pool (also referred to as "virtual funds" by Azuro) refers to using one or more liquidity pools to back multiple prediction markets rather than having a dedicated pool for each market. The system only locks the amount of liquidity needed to cover the maximum possible payout for each market, recalculating this amount with each new bet. This virtual liquidity accounting adjusts automatically as bets are placed, effectively acting as an automated market maker for betting odds.

BENEFITS	RISKS
• **Lower Capital Requirements**: Liquidity providers can support multiple markets with a single capital contribution. • **Efficient Liquidity Usage**: Using a global pool, the system can support a more significant number of markets with the same total liquidity.	• **Potential for Overextension**: If not properly managed, the system could allocate more virtual funds than actual liquidity. • **Interdependence Risk**: Issues in one market could potentially affect liquidity across the entire system.

• **Flexible Market Creation**: The platform can easily create new markets without sourcing new liquidity.	

Logarithmic Market Scoring Rule (LMSR)

Invented by prediction market pioneer Robert Hanson (2002) and implemented by Gnosis (2019), LMSR is an automated market maker mechanism for prediction markets that uses a cost function to determine the price of outcome shares based on the number of shares for each outcome and a liquidity parameter.

BENEFITS	RISKS
• **Continuous liquidity**: LMSR ensures a price for every outcome at all times, allowing traders to buy or sell without needing counterparties. This constant liquidity availability encourages more frequent trading and potentially increases overall market depth. • **Informed incentives**: The mechanism encourages	• **High initial funding**: LMSR often requires substantial initial funding, especially for markets with many outcomes, which can be a barrier to entry. • **Price volatility**: Small trades can cause significant price movements in markets with low liquidity, potentially deterring some

early and informed trading, potentially leading to more accurate predictions sooner. This can attract more participants and increase liquidity as the market matures.

- **Outcome flexibility**: LMSR can theoretically handle markets with infinite outcomes, making it versatile for various prediction scenarios. This flexibility allows for creating more diverse markets, potentially attracting a more comprehensive range of participants and increasing overall liquidity across the platform.
- **Predictable risk**: The maximum loss for the market maker is known in advance, calculated as b * log(n), where b is the liquidity parameter and n is the number of outcomes. This predictability can traders.
- **Parameter sensitivity**: The liquidity parameter needs precise adjustment to balance price sensitivity and funding requirements, which can be challenging.

encourage more liquidity providers to participate, as they can better manage their risk exposure.	

Pythagorean Bonding Curve AMM

Developed by Obyte, the Pythagorean bonding curve AMM uses a mathematical formula inspired by the Pythagorean theorem to manage liquidity and price tokens in prediction markets while addressing some limitations of other liquidity solutions such as LMSR. An autonomous agent (AA) issues YES, NO, and optionally DRAW tokens for each market, with token prices determined by their relative supplies. Users can buy or sell tokens anytime, adjusting prices automatically as a function of each token's supply. Token prices are calculated as partial derivatives of the reserve concerning token supply, ensuring that the total value of all tokens equals the reserve. After an event, holders of winning tokens can redeem their share of the total reserve, while other tokens become worthless. Trading fees are incorporated by adjusting the coefficient, which increases all token prices and allows every token holder to benefit from accumulated fees.

BENEFITS	RISKS
• **Constant Liquidity**: The bonding curve ensures that tokens can be bought or sold at any time,	• **Price Sensitivity**: Large trades can significantly impact token prices, potentially leading to

solving the liquidity bootstrapping problem faced by traditional prediction markets.

- **Fair Distribution**: The mathematical model ensures that profits and losses are distributed relatively among participants based on their contributions.
- **Scalability**: The system can handle many markets and participants without requiring separate liquidity pools for each market.
- **Flexible Liquidity Provision**: Anyone can add liquidity at any time by buying tokens, and liquidity is never lost to bettors.
- **Incentivized Participation**: Every token holder becomes a liquidity provider as their tokens appreciate due to accumulated trading fees.

slippage for traders.

- **Non-Intuitive Pricing**: Unlike other prediction market models, token prices don't directly represent probabilities, which may confuse some users.
- **Initial Liquidity Requirements**: While lower than some alternatives, the system still requires some initial liquidity to function effectively.

Conclusion

As these mechanisms evolve, prediction markets stand a chance to transform crypto's speculative madness into a precision instrument for forecasting matters of public interest. This would open the door to a range of applications that prediction market nerds have prophesied for decades, including:

- Incentivized whistleblowing & intel gathering
- Decision markets for corporate governance and public policy
- Futarchy (also see MetaDAO)
- Hedging information security risks and other risk like weather events and economic crises
- Parametric insurance
- Funding science and academic research

In this way, prediction markets can move beyond predicting the future to actually facilitating coordination that improves the future.

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